01. Foundational Concepts of Analytics

1. Business Analytics - refers to the process of analyzing business performance and assisting with decision-making through the use of data, statistical techniques and technologies.

* Real world Example

- A retail company uses business analytics to study customer purchase data. By analyzing buying patterns, the company can predict which products will sell during certain season and adjust inventory levels accordingly, this helps reduce stock shortages and improve sales.

2. Data Analytics - is the process of analyzing, cleaning, manipulating and interpreting data in order to find patterns, find relevant information and support ~~des~~ decision-making. It helps businesses in transforming raw data into useful insights.

Real world example

→ a bank uses data analytics to monitor customer ~~transform~~ transactions and detect unsual activity. if a transation looks ~~sp~~ suspicious the system flags it for review, helping prevent foud

3.

* Business intelligence (BI) - refers to the methods techniques and technology used to gather, examine and display corporate data in a way that facilitates improved decision making.

Real world example

* A supermarket chain uses BI tools to create dashboards that show daily sales by product and store location. managers can quickly see which products are underperforming and adjust pricing or promotions to increase sales

4
Data Driven Decision making - is the process of making business decisions based on data analysis and evidence rather than gut feeling or conjecture.
Real world example
- An e-commerce company analyzes website data to decide which products to feature on its homepage. by promoting items that data shows are more likely to sell. the company increase conversations and revenue.

5
Big Data - refers to extremely large and complicated datasets that are difficult to handle with conventional data processing methods
Real world example
- A streaming company like Netflix uses big data to analyze millions of user interactions. this data helps recommend personalized content to users, improving user engagement and reten tion.

6.
- Descriptive Analytics - is the process of analyzing historical data to comprehend past events. it helps companies to monitor performance and spot trends by ~~con~~ summarizing raw data into ~~insight~~

insightful reports, dashboards and visualizations

Examples

- A retail store uses descriptive analytics to generate monthly sales reports. by reviewing these reports, managers can see which Product sold the most, which stores performed best and how sales changed over time.

7

* Diagnostic Analytics - is the process of analyzing historical data to ~~at~~ comprehend past events.

Example

- A retail company notices a sudden drop in sales using diagnostic, analytics, they analyze data by region, Product category and customer segment. they discover that a competitor launched a discount campaign in a key region, which caused the decline in their sales.

8.

* Predictive Aanalytics - is a kind of data analysis that forecasts future patterns or outcomes using statistical algorithms, machine ~~learing~~ learning techniques and previous data. It provides an answer to the question: what is likely to happen nex?.

Example:

- A bank uses predictive analytics to assess wether a customer is likey to default on a loan. By analyzing past customer data, the bank can predict risk levels and make better lending decisions.

9.

* Prescriptive analytics - is a type of data analytics

that makes recommendations for how to accom plish specific goals. it answer the question "what should we do next", using methods like optimization, simulation and machine learning in addition to insights from descriptive and predictive analytics

* <u>Data literacy</u> – is the capacity to successfully read, understand, interpret and ~~comu~~ communicates data efficitively. it encompasses ~~abb~~ abilities like data analysis, insight drawing and using Data to make well informed decisions in a business or everyday context.

<u>example</u>

* A marketing team uses data literacy skills to interpret campaign perfomance dashboards, by understanding metrics like click through rates and conversion rates, they can adjust their strategy to improve future marketing results.

<u>02: Data, Measurement & Reporting</u>

11. <u>Structured Data</u> – refer to data that is ~~that is~~ arranged in a specific manner, usually in rows and columns (such as tables or spreadsheet)

<u>example</u>

A supermarket uses structured data in its sales database, where each transaction is recorded in a table with fields, such as product ID, price, quantity and date

12 <u>unstructured data</u> - refers to data that lacks a set format or structure, like a table's rows and columns. Emails, social media postings, photos, audio, and videos are among the forms that frequently contain a lot of text or multimedia content.

<u>example</u>

* A customer service department analyze unstructured data from customer emails and chat messages to identify common complaints. By using text analytics the company can detect patterns such as repeated issues with delivery delays and improve services quality B.

* <u>Dat warehousing</u> - is a centralized system for storing substantial amonts of semi-structered and structured data from various sources.

<u>examples</u>

A retail company uses a data warehouse to combine sales from physical stores, online platforms and customer loyalty ~~sten~~ systems.

14

<u>Data mining</u> - is the process of applying methods like machine learning, statscs, and database systems to extract patterns, trends and helpful information from massive datasets. It helps business in trans ~~for~~ forming upprocessed data into insightful information for decision making

<u>example</u>

* A supermarket uses data mining to analyze customer purchase histories and discovers that people who buy diapers often also baby wipes. Based on this insight

the Store can place these items together or create targeted promotions to increase sales is

Data visualization - is the grathical display of data utilizing visual components including dashboards graphs, charts, and maps, By clarifying patterns, trends, ~~and~~ it makes complex data easier for individuals to understand.

example

* A sales manager uses a dashboard with bar charts and line graphs to visualize monthly sales performance across different regions. this makes it easy to quickly identify which regions are performing well and which need improvement

16

* Dashboard - is visual interface that uses tables, graphs and charts to presents important metrics, indicators and data summaries in one place.

example

- A logistics company uses a dashboard to track deliveries, ~~vehicl~~ vehicle locations, fuel usage and delivery times, this helps managers quickly identify delays and optimize delivery operations.

Metric - is a measurable metric used to monitor, analyze and assess how well a certain business process, activity or objective is performing

example

An e-commerce company tracks the conversion rate as a metric, which measure the percentage of website ~~visis~~ visitors who complete a purchase. If the

the conversion rate drops, the business may investigate issues with website design or pricing

18

* <u>Key performance</u> indicator (KPI) – is a quantifiable indicator of how well a team, organization or individual is accomplishing important business goals. KPI are used to assess how well strategic objectives are beign met.

<u>example</u>:

* A call centre uses "average response time" as a KPI to measure how quickly agents answer customer calls. If response time are too high, management may hire more staff or improve training to improve service efficiency.

19.

* <u>Benchmark</u> – is a benchmark or point of reference for evaluating performance, procedures or results in relation to industry norms, competitors or best practices. It assist organization in assessing their performance and pinpointing areas in need of development.

<u>examples</u>

* A mobile network company compares it customers satisfaction score against industry benchmarks. If its score is lower than competitors, it may improve network coverage or customer service to remain competitive.

20

* <u>Segmentation</u> – is the practice of breaking up a big market, client base or set of data into smaller, more distinct groups according

to shared traits like preferences, behavior or demographics. ~~It~~ It enable companies to more successfully target each demographic with customized ~~tra~~ tactics
Example
* A clothing retailer uses customer segmentation to group shoppers by age and buying behavior for example it targets young adults with trendy fashion ads while promoting formal wear to working professionals.

03: Customer & Sales KPIs

21. <u>Conversion Rate</u> - A metric that calculates the percentage of users or visitors who complete a desired action ~~to the total number of visits~~ relative to the total number of visits. It is frequently used in company marketing and sales to assess how well a website, campaign or advertisement converts prospective clients into actual clients

formular:

$$\text{Conversion Rate} = \frac{\text{Number of Conversion}}{\text{Total Visitors}} \times 100$$

<u>Example</u>

An online store receives 1000 website visitors in a day an 50 of them make a purchase. the conversion rate is 5%, the business uses the KPI to evaluate the effectiveness of its website design and ~~market~~ marketing campaigns.

22 <u>Customer Acquisition Cost (CAC)</u> - is a business metric that calculates how much it cost a business to bring on a new client. it is calculated by dividing marketing, advertising, sales force salaries and other related costs by the number of new clients acquired over a given time frame.

$$CAC = \frac{\text{Total Sales and Marketing Costs}}{\text{Number of New Customers acquired}}$$

<u>Example</u>

A Software-as-a service (SaaS) company spends R100,000 on marketing and sales in one month and acquire 200 new customers. the CAC is R500 per customer. The company uses this metric to evaluate whether its marketing efforts are cost effective

23

* <u>Customer lifetime value (CLV/LTV)</u> - ~~is a measure that~~ is a metric that estimates the total revenue a business can anticipate to earn from a client over the course of their partnership

examples

* A subscription base streaming service calculates that an average customer stays ~~subscrib~~ subscribed for 3 years and generates R1,200 per year. The CLV is R3,600. this helps the company decide how much it can afford to spend on acquiring new subscribes.

24

* Churn Rate - is a business indicator that calculate the proportion of consumers that discontinue using a good or service over a given time frame.

$$\text{Churn Rate} = \frac{\text{Customers lost During Period}}{\text{Total Customers at Start of Period}} \times 100$$

Example

- A mobile network company starts the month with 10 000 subscribers and loses 500 subscribers by the end of the month. The churn rate is 5%, the company uses this information to improve customer services and reduce cancellations

25

* Customer Retention Rate - is a metric that calculates the percentage of clients that a business keeps over a given time frame

$$\text{Customer Retention Rate} = \frac{\text{Customers at End of Period} - \text{New Customers}}{\text{Customer at Start of Period}}$$

example

* A gym start the year with 1000 members, gains 200 new members and ends the year with 1050 members. the high retention rate shows that most members continue their subscriptions, helping the gym ~~maintin~~ maintain stable revenue

26

Net Promoter Score (NPS) - is a customer loyalty indicator that measures the likelihood that customers ~~or~~ will tell others about a company's goods or services.

it is predicated on answers to a survey question scored between 0 and 10: How likely are you to suggest us?

NPS = % Promoters − % Detractors

Examples

- A telecom Company survey Customers after service interations. if many Customers rate their experience highly and recommend the services the Company gets a high NPS, indicating strong Customer ~~Satisfi~~ Satisfaction and loyalty

27

Average order value (AOV) - is a measure of how ~~money~~ much money a Costomer spends on each order or transaction. It helps Company in Comprehending Costomer behavior and assessing the sucess of marketing, pricing and upselling tactics.

$$AOV = \frac{\text{Total Revene}}{\text{number of orders}}$$

Example

* An online Clothing store earns R50 000 from 1000 in a month. The AOV is R50. the business uses this insight to encourage Customer to buy more per order by offering bundle deals or free shipping thresholds.

28

Sales funnel - is a marketing and sales models that represents the customer's journey from intial awareness of a product or service to making the ultimate purchase.

example

* A online eduction platform attracts visitors through social media ads. Some of these visitors sign up for free trial courses and a ~~Sime~~ smaller group eventually pays for full certification programs. The company uses the sales funnel to track where customers drop off and improve conversions

04: Marketing & Campaign Analytics

29. Marketing Campaign - is a coordinated set of events and messaging intended to advertise a product, service or brand to a particular market over a predetermined time frame

example

- A smartphone company launches a marketing campaign for a new phone model using TV ads instogram influencers and email promotion. The goal is to create awareness and encourage customers to pre-order the device.

30

* Post-campaign analysis - is the process of evaluating and assessing a marketing campaign after it has concluded in order to compare its effectivenes ss to predetermined goals.

To identify what worked well and what has to be improved for upcoming campaigns, it entails analyzing important data including reach, engagement, ~~conversat~~ conversions and return on investment (ROI)

<u>example</u>

* After running a social media advertising campaign, a retail company reviews the results and finds that instagram ads generated more sales than facebook ads.

31

* <u>Incremental Revenue</u> - is the extra money made from a particular business plan, campaign or activity over what would have been made in the absence of such action. It is employed to quantify the immediate financial effects of advertising campaigns, promotions or commercial choices.

<u>examples</u>

* A supermarket runs a "buy one get on free" promotion on snacks and notices an extra R20 000 in sales compared to a normal week. That additional R20 000 is the incremental revenue generated by the promotion

32.

* <u>Return on investment (ROI)</u> - a financial measure used to assess an investment's efficiency or profitability. It is typically represented as a percentage and calculate the return on investment in relation to the investment cost

$$ROI = \frac{\text{Net profit}}{\text{cost of investment}} \times 100$$

33. Return on Ad Spend (ROAS) - is a marketing metric that calculates the amounts of money made for each advertising unit. It demonstrates how successful an advertising effort is at making sales.

$$ROAS = \frac{\text{Revenue from Ads}}{\text{Cost of Ads}}$$

example

an online fashion store spends R 5000 on a facebook ad campaign and generates R25000 in sales from that campaign. The ROAS is 5:1 meaning for every R1 spent on ads, the business earns R5 in revenue.

34

*Click-Through Rate (CTR) - is a metric used in digital marketing that calculates the proportion of viewers who click on a call-to-action, link or advertisement. It is used to assess how well a piece of information or advertisement draws in users

$$CTR = \frac{\text{Click}}{\text{Impressions}} \times 100$$

Example

*An e-commerce company runs an online banner ad that is shown 10000 times and receives 300 clicks. The CTR is 3%. The company uses this metric to determine whether the ad design and message are effective.

35 Bounce Rate → is the percentage of visitors who arrive at the website and depart without doing anything more, such as clicking on another page, completing a form or making a purchase

example

- an e-commerce website gets 2000 visitors to a product page, but 1200 leave without clicking anything else. the bounce rate is 60%, which may indicate the page is not engaging or clear enough

36

A/B testing - is a technique for comparing two versions of a website, application, or advertisement to determine which one works better

Example

*A online store tests two checkout buttons
version A: "Buy now" (green button)
version B: "Add to cart" (blue button)
after testing version B gets more clicks so the business adopt it

37

Lead generation - is the practice of gathering potential ~~cile~~ clients contact details or showing interest in a product or service in order to draw in and covert them

example

A software company offers a free ebook on data ~~AGA~~ ~~analysis~~ analytics. users must

enter their email to download it. those emails sign-ups are considered leads.

38

<u>Marketing Attribution</u> - is the act of determing which marketing channels or touchpoints - such as advertisements, ~~email~~ emails and social medial helped a ~~custome~~ customer make a purchase

<u>example</u>

A customer ~~sess~~ sees a facebook ad, then later clicks a google ad and makes a purchase, marketing attribution helps the company determine which channel influenced a sale.

Reference

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